APPENDICES

(1) Properties of Trigonometric Functions:

A-1

$$\tan x = \frac{\sin x}{\cos x}$$

$$\cot x = \frac{1}{\tan x} = \frac{\cos x}{\sin x}$$

$$\sec x = \frac{1}{\cos x}$$

$$\csc x = \frac{1}{\sin x}$$

A-2

$$\sin(-x) = -\sin x$$

$$\cos(-x) = \cos x$$

$$\tan(-x) = -\tan x$$

A-3

$$\sin^2 x + \cos^2 x = 1$$

$$\tan^2 x + 1 = \sec^2 x$$

$$\cot^2 x + 1 = \csc^2 x$$

A-4

$$\sin(x \pm y) = \sin x \cos y \pm \cos x \sin y$$

$$\cos(x \pm y) = \cos x \cos y \mp \sin x \sin y$$

$$\tan(x \pm y) = \frac{\tan x \pm \tan y}{1 \mp \tan x \tan y}$$

A-5

$$\sin 2x = 2\sin x \cos x$$

$$\cos 2x = \cos^2 x - \sin^2 x$$

$$\tan 2x = \frac{2\tan x}{1 - \tan^2 x}$$

A-6

$$\sin x \cos y = \frac{1}{2}\left[\sin(x+y)+\sin(x-y)\right]$$

$$\cos x \sin y = \frac{1}{2}\left[\sin(x+y)-\sin(x-y)\right]$$

$$\cos x \cos y = \frac{1}{2}\left[\cos(x+y)+\cos(x-y)\right]$$

$$\sin x \sin y = \frac{1}{2}\left[\cos(x-y)-\cos(x+y)\right]$$

A-7

$$\sin^2 x = \frac{1}{2}(1-\cos 2x)$$

$$\cos^2 x = \frac{1}{2}(1+\cos 2x)$$

A-8

$$\sin^3 x = \frac{1}{4}(3\sin x - \sin 3x)$$

$$\cos^3 x = \frac{1}{4}(3\cos x + \cos 3x)$$

(2) Properties of Hyperbolic Functions:

B-1

$$\sinh x = \frac{1}{2}\left(e^x - e^{-x}\right)$$

$$\cosh x = \frac{1}{2}\left(e^x + e^{-x}\right)$$

$$\tanh x = \frac{\sinh x}{\cosh x}$$

$$\coth x = \frac{1}{\tanh x} = \frac{\cosh x}{\sinh x}$$

$$\operatorname{sech} x = \frac{1}{\cosh x}$$

$$\operatorname{csch} x = \frac{1}{\sinh x}$$

B-2

$$\sinh(-x) = -\sinh x$$

$$\cosh(-x) = \cosh x$$

$$\tanh(-x) = -\tanh x$$

B-3

$$\cosh^2 x - \sinh^2 x = 1$$

$$1 - \tanh^2 x = \operatorname{sech}^2 x$$

$$\coth^2 x - 1 = \operatorname{csch}^2 x$$

B-4

$$\sinh(x \pm y) = \sinh x \cosh y \pm \cosh x \sinh y$$

$$\cosh(x \pm y) = \cosh x \cosh y \pm \sinh x \sinh y$$

$$\tanh(x \pm y) = \frac{\tanh x \pm \tanh y}{1 \pm \tanh x \tanh y}$$

B-5

$$\sinh 2x = 2\sinh x \cosh x$$

$$\cosh 2x = \cosh^2 x + \sinh^2 x$$

$$\tanh 2x = \frac{2\tanh x}{1 + \tanh^2 x}$$

B-6

$$\sinh x \cosh y = \frac{1}{2}\left[\sinh(x + y) + \sinh(x - y)\right]$$

$$\cosh x \sinh y = \frac{1}{2}\left[\sinh(x + y) - \sinh(x - y)\right]$$

$$\cosh x \cosh y = \frac{1}{2}\left[\cosh(x + y) + \cosh(x - y)\right]$$

$$\sinh x \sinh y = \frac{1}{2}\left[\cosh(x + y) - \cosh(x - y)\right]$$

B-7

$$\sinh^2 x = \frac{1}{2}(\cosh 2x - 1)$$

$$\cosh^2 x = \frac{1}{2}(\cosh 2x + 1)$$

B-8

$$\sinh^3 x = \frac{1}{4}\left(\sinh 3x - 3\sinh x\right)$$

$$\cosh^3 x = \frac{1}{4}\left(\cosh 3x + 3\cosh x\right)$$

(3) Properties of Exponential Function:

If $n, m \in R,\ a > 0,\ b > 0$

(1) $a^m a^n = a^{m+n}$

(2) $\dfrac{a^m}{a^n} = a^{m-n}$

(3) $\left(a^m\right)^n = a^{mn}$

(4) $\left(ab\right)^n = a^n b^n$

(5) $\left(\dfrac{a}{b}\right)^n = \dfrac{a^n}{b^n}$

(6) $a^{-n} = \dfrac{1}{a^n}$

(4) Properties of Logarithmic Function:

If $a, b, c \in R,\ a > 0,\ b > 0,\ c > 0,\ c \neq 1$

(1) $\log_c ab = \log_c a + \log_c b$

(2) $\log_c \dfrac{a}{b} = \log_c a - \log_c b$

(3) $\log_c a^n = n \log_c a, \qquad n \in R$

(4) $\log_c \sqrt[n]{a} = \dfrac{1}{n}\log_c a, \qquad n \in R$

(5) $\log_c c = 1,\ \log_c 1 = 0$

(6) $\log_{10} c = \log c$ Usual logarithm

(7) $\log_e c = \ln c,\ e \approx 2.71828$ Natural logarithm

(5) Basic Differentiation Formulas:

(1) $\frac{d}{dx}C = 0, \quad C \in R$

if $u = g(x)$ *is a differentiable function of* x *,then*

(2) $\frac{d}{dx}u^n = nu^{n-1}\frac{du}{dx}, \quad n \in R$

(3) $\frac{d}{dx}e^u = e^u \frac{du}{dx}$

(4) $\frac{d}{dx}a^u = a^u \ln a \frac{du}{dx}$

(5) $\frac{d}{dx}\ln u = \frac{1}{u}\frac{du}{dx}$

(6) $\frac{d}{dx}\log_a u = \frac{1}{u \ln a}\frac{du}{dx}$

(7) $\frac{d}{dx}\sin u = \cos u \frac{du}{dx}$

(8) $\frac{d}{dx}\cos u = -\sin u \frac{du}{dx}$

(9) $\frac{d}{dx}\tan u = \sec^2 u \frac{du}{dx}$

(10) $\frac{d}{dx}\cot u = -\csc^2 u \frac{du}{dx}$

(11) $\frac{d}{dx}\sec u = \sec u \tan u \frac{du}{dx}$

(12) $\frac{d}{dx}\csc u = -\csc u \cot u \frac{du}{dx}$

(13) $\frac{d}{dx}\sinh u = \cosh u \frac{du}{dx}$

(14) $\frac{d}{dx}\cosh u = \sinh u \frac{du}{dx}$

(15) $\frac{d}{dx}\tanh u = \operatorname{sech}^2 u \frac{du}{dx}$

(16) $\frac{d}{dx}\coth u = -\operatorname{csch}^2 u \frac{du}{dx}$

(17) $\frac{d}{dx}\text{sech}u = -\text{sech}u\tanh u\frac{du}{dx}$

(18) $\frac{d}{dx}\text{csch}u = -\text{csc}\,\text{h}u\coth u\frac{du}{dx}$

(19) $\frac{d}{dx}\sin^{-1}u = \frac{1}{\sqrt{1-u^2}}\frac{du}{dx}, \quad |u|<1$

(20) $\frac{d}{dx}\cos^{-1}u = \frac{-1}{\sqrt{1-u^2}}\frac{du}{dx}, \quad |u|<1$

(21) $\frac{d}{dx}\tan^{-1}u = \frac{1}{1+u^2}\frac{du}{dx}$

(22) $\frac{d}{dx}\cot^{-1}u = \frac{-1}{1+u^2}\frac{du}{dx}$

(23) $\frac{d}{dx}\sec^{-1}u = \frac{1}{u\sqrt{u^2-1}}\frac{du}{dx}, \quad |u|>1$

(24) $\frac{d}{dx}\csc^{-1}u = \frac{-1}{|u|\sqrt{u^2-1}}\frac{du}{dx}, \quad |u|>1$

(25) $\frac{d}{dx}\sinh^{-1}u = \frac{1}{\sqrt{1+u^2}}\frac{du}{dx}$

(26) $\frac{d}{dx}\cosh^{-1}u = \frac{1}{\sqrt{u^2-1}}\frac{du}{dx}, \quad |u|>1$

(27) $\frac{d}{dx}\tanh^{-1}u = \frac{1}{1-u^2}\frac{du}{dx}, \quad |u|<1$

(28) $\frac{d}{dx}\coth^{-1}u = \frac{1}{1-u^2}\frac{du}{dx}, \quad |u|>1$

(29) $\frac{d}{dx}\text{sech}^{-1}u = \frac{-1}{u\sqrt{1-u^2}}\frac{du}{dx}, \quad 0<u<1$

(30) $\frac{d}{dx}\text{csch}^{-1}u = \frac{-1}{|u|\sqrt{1+u^2}}\frac{du}{dx}, \quad |u|\neq 0$

*(6) **Basic Integral Formulas:***

(1) $\int du = u + C$

(2) $\int u^n\,du = \frac{u^{n+1}}{n+1} + C$

(3) $\int \frac{1}{\sqrt{u}} du = 2\sqrt{u} + C$

(4) $\int \frac{1}{u} du = \ln|u| + C$

(5) $\int e^u \, du = e^u + C$

(6) $\int a^u \, du = \frac{a^u}{\ln a} + C, \qquad a > 0, a \neq 1$

(7) $\int \sin u \, du = -\cos u + C$

(8) $\int \cos u \, du = \sin u + C$

(9) $\int \tan u \, du = \ln|\sec u| + C$

(10) $\int \cot u \, du = \ln|\sin u| + C$

(11) $\int \sec u \, du = \ln|\sec u + \tan u| + C$

(12) $\int \csc u \, du = \ln|\csc u - \cot u| + C$

(13) $\int \sec^2 u \, du = \tan u + C$

(14) $\int \csc^2 u \, du = -\cot u + C$

(15) $\int \sec u \tan u \, du = \sec u + C$

(16) $\int \csc u \cot u \, du = -\csc u + C$

(17) $\int \sinh u \, du = \cosh u + C$

(18) $\int \cosh u \tan u \, du = \sinh u + C$

(19) $\int \tanh u \, du = \ln|\cosh u| + C$

(20) $\int \coth u \, du = \ln|\sinh u| + C$

(21) $\int \operatorname{sech} u \, du = \tan^{-1}(\sinh u) + C$

(22) $\int \operatorname{csch} u \, du = \ln\left|\tanh \frac{1}{2} u\right| + C$

(23) $\int \operatorname{sech}^2 u \, du = \tanh u + C$

(24) $\int \operatorname{csch}^2 u \, du = -\coth u + C$

(25) $\int \operatorname{sech} u \tanh u \, du = -\operatorname{sech} u + C$

(26) $\int \operatorname{cschu}\coth u \; du = -\operatorname{cschu} + C$

(27) $\int \frac{1}{\sqrt{a^2 - u^2}}\, du = \sin^{-1}\left(\frac{u}{a}\right) + C, \qquad u^2 < a^2$

(28) $\int \frac{1}{\sqrt{a^2 + u^2}}\, du = \sinh^{-1}\left(\frac{u}{a}\right) + C, \qquad u > 0$

(29) $\int \frac{1}{\sqrt{u^2 - a^2}}\, du = \cosh^{-1}\left(\frac{u}{a}\right) + C, \qquad u > a > 0$

(30) $\int \frac{1}{a^2 + u^2}\, du = \frac{1}{a}\tan^{-1}\left(\frac{u}{a}\right) + C$

(31) $\int \frac{1}{a^2 - u^2}\, du = \begin{cases} \frac{1}{a}\tanh^{-1}\left(\frac{u}{a}\right) + C, & u^2 < a^2 \\ \frac{1}{a}\coth^{-1}\left(\frac{u}{a}\right) + C, & u^2 > a^2 \end{cases}$

(32) $\int \frac{1}{u\sqrt{u^2 - a^2}}\, du = \frac{1}{a}\sec^{-1}\left(\frac{u}{a}\right) + C, \qquad u^2 > a^2$

(33) $\int \frac{1}{u\sqrt{a^2 - u^2}}\, du = -\frac{1}{a}\operatorname{sech}^{-1}\left(\frac{u}{a}\right) + C, \qquad 0 < u < a$

(34) $\int \frac{1}{u\sqrt{a^2 + u^2}}\, du = -\frac{1}{a}\operatorname{csch}^{-1}\left(\frac{u}{a}\right) + C, \qquad u \neq 0$

(35) $\int u \; dv = uv - \int v \; du$